

COST OF OPPORTUNITY PROBLEM SET—SUGGESTED SOLUTIONS

ECON101—INTRODUCTION TO MICROECONOMICS.

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Problem 1

Students could be aware that there are other, more important uses of the limited budget that NU has. Remember, scarcity means that having more of one thing necessarily means having less of another. If USM spends 9.5 billion tenge in a new recreation center/food court, then it will have 9.5 billion less to spend on other projects, like improving the conditions of student dormitories. If the value to students of those alternative uses is greater than the value they place on the new recreation center/food court, then they would have voted “no”.

Of course, the analysis above relies on assuming that these students understand the concept of *scarcity* and the *economic problem*. It is thus highly likely that these students paid attention in their ECON101 class.

Problem 2

Remember that we have the following:

$$\text{Economic surplus of activity } x = \text{Benefit}(x) - \text{Cost}(x).$$

Remember that costs should include both explicit and implicit costs, i.e., it should be a complete and correct measure of the cost of opportunity. Let x = Buying the Finance Bro outfit. There are two ways to proceed. Let N be the number of friends in Beksultan’s Finance Bro circle, and M the number of students outside that circle.

- (a) The implicit cost of being loathsome to other students enters (is added) as an implicit cost in the opportunity cost calculation.
- $B(x, N)$ = the monetary value that Beksultan attaches to impressing N students,
 - $C(x, M)$ = the total cost of the outfit + the monetary value lost due to M students finding Beksultan loathsome.
 - Thus surplus is $S(x, N, M) = B(x, N) - C(x, M)$.
- (b) The implicit cost of being loathsome to other students enters (is subtracted) as an implicit cost in the benefit calculation.
- $B(x, N, M)$ = the monetary value that Beksultan attaches to impressing N students – the monetary value lost due to M students finding Beksultan loathsome.
 - $C(x)$ = the total cost of the outfit
 - Thus surplus is $S(x, N, M) = B(x, N, M) - C(x)$.

Problem 3

In this version of the problem, Beksultan can no longer impress his N friends from the Finance Bro Circle because all of them are wearing the outfit. It is easier to see what happens using method (a) above: the benefit that Beksultan derives from buying the outfit is now $B(x, N) = 0$. This implies that $S(x, N, M) < 0$. Thus, it is clear that he should not waste his money (or his life) trying to impress others.

Problem 4

Consider the following table:

Kg of Compost	Kg of tomatoes	Revenue	Cost	Profit	Marginal Revenue	Marginal Cost
0	100	150000	0	150000	150000	0
1	120	180000	1850	178150	30000	1850
2	125	187500	3700	183800	7500	1850
3	128	192000	5550	186450	4500	1850
4	130	195000	7400	187600	3000	1850
5	131	196500	9250	187250	1500	1850
6	131.5	197250	11100	186150	750	1850

From the table above it is clear that Kamilla should use 4 kilograms of compost, since that is where profit is maximized. Also note that at 4 kilograms, marginal revenue is greater than marginal cost, and that by increasing by one more kilogram of compost, marginal revenue becomes lower than marginal cost. This is the *cost-benefit principle*.

Problem 5

If Daniyar kept the 200,000 tenge and invested it in additional mushrooms, at the end of a year's time he would have 400,000 tenge worth of mushrooms to sell. Therefore, Bagdat must give Daniyar 200,000 tenge in interest in order for Daniyar not to lose money on the loan.

Problem 6

Even though you earned four times as many points from the first question than from the second, the last few seconds you spent on question 2 added more points to your score than did the last few seconds you spent on question 1. This suggests that if you spent a little more time on question 2 and a little less time on question 1, then you'd get more extra points on question 2 than you'd lose on question 1.

Problem 7

According to our discussion in the lectures, specifically, the *cost-benefit principle*, both Akmaral and Medina should make the same decision. After all, the benefit of seeing the movie is the same in both cases, and the cost of seeing the play—at the moment each must decide—is exactly 1,600 tenge. Do not make the mistake of thinking that in the case of the lost ticket, the cost of seeing the movie is not

1,600 but 3,200, i.e., the price of two tickets. This is an example of a *sunk cost*. The loss of a ticket is not different from the loss of a 1,600 tenge, as we saw in one of the examples from the lectures. So in each case, the question is whether seeing the movie is worth spending 1,600 tenge. If it is worth it, both Akmaral and Medina should see the movie; otherwise they should not enter the cinema. Whichever you answer, it must be the same for both Akmaral and Medina.

Problem 8

If Nazima follows the principles we have studied (and rational people tend to do it), she continues each scooter ride as long as the marginal benefit of riding a little longer is greater than the marginal cost. Her usage under the current plan tells us that Nazima values the tenth minute of each ride at no less than its marginal cost of 200 tenge. For a 10-minute ride, both plans charge exactly the same amount, 2,000 tenge. But at that point, under the new plan, the marginal cost is only 100 tenge per minute. Since her benefit of riding an additional minute is the same under both plans, Nazima is likely to take longer rides under the new plan.

Problem 9

In Plan A, the marginal extra cost for each additional kilogram of food is 0 tenge, so everybody will keep eating until the extra benefit from eating an extra kilogram of food is also equal to 0. In plan B, however, the cost of eating an extra kilogram of food is 2,000 tenge for those eating more than 100 kilograms. Students that are eating more than 100 kilograms of food will stop eating when the benefit of eating an extra kilogram falls to 2,000 tenge.

For students that are above the average consumption level of 100 kilograms, Plan A makes it more likely that they will keep eating, compared to Plan B. Thus, food consumption will be higher under plan A.